

Application of tensorOmics methods to longitudinal omics data: trajectory-based analysis of aging and vaccine response

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1 Introduction

Longitudinal omics studies follow the same individuals at several time points. Instead of taking only one biological measurement, they allow us to observe how molecular profiles change over time. This type of design is useful for studying processes such as aging, infection, vaccination or treatment response.

However, these data are not easy to analyse. In omics studies, thousands of genes or probes can be measured, but the number of individuals is often much smaller. In addition, each individual is measured several times. This means that the data are not just a simple table: they contain both molecular information and temporal information.

A common approach is to transform the data into a matrix and then apply methods such as PCA or PLS-DA. This can be useful for a first analysis, but it can also lose part of the longitudinal structure. For example, if samples collected from the same person at different days are treated as independent samples, the model does not fully use the fact that they belong to the same individual.

Tensor-based methods provide a more natural way to organise this type of data. A longitudinal transcriptomic dataset can be represented as:

$$\text{subjects} \times \text{features} \times \text{time}$$

In this structure, each subject is kept as a complete trajectory. The tensorOmics framework extends this idea to several methods, including tensor PCA (tPCA) and tensor PLS-DA (tPLS-DA), in order to analyse longitudinal omics data without completely flattening the time dimension.

The aim of this report was not to develop a new statistical method, but to apply tensorOmics methods to real public datasets and evaluate what kind of information they can reveal. I focused on two longitudinal transcriptomic examples. The first analysis used tPLS-DA on a fish aging RNA-seq dataset to study whether early transcriptomic trajectories could be associated with future lifespan. The second analysis used tPCA on an independent human vaccination dataset from GEO to test whether tensor-based exploration could reveal age-associated transcriptional trajectories after vaccination.

2 Biological questions and datasets

2.1 Fish aging transcriptomic dataset

The first dataset was GSE150318, a longitudinal RNA-seq study of fin biopsies from the short-lived killifish *Nothobranchius furzeri* [3]. This species is often used in aging research because it has a naturally short lifespan. The biological question was whether early transcriptomic profiles could contain information related to future lifespan.

The original study measured fin biopsies from 114 animals at 10 and 20 weeks of age. In my analysis, after keeping individuals with complete information and filtering the expression matrix, I used 113 fish, 1000 variable genes and two time points. The fish were divided into two lifespan groups: 59 short-lived individuals and 54 long-lived individuals. I also kept the real age at death, because it allowed me to test whether the components were related to lifespan as a continuous variable, not only to a binary label.

2.2 Human vaccine response dataset

For the second analysis, I searched GEO for a dataset that was longitudinal, publicly available, biologically interpretable, and suitable for tensorOmics.

I selected GSE79396, a study of immune responses after Herpes zoster vaccination with Zostavax in young and elderly healthy adults [4]. The full study included 77 subjects enrolled in Atlanta and Denver, with blood collected at baseline and multiple post-vaccination time points. The original Cell article used transcriptomics, metabolomics, immune cell populations and cytokines to describe age-related immune differences and vaccine response signatures.

This dataset was suitable for several reasons. First, it had repeated measurements on the same subjects. Second, the early time points after vaccination were biologically meaningful, because transcriptional responses to vaccines often occur rapidly. Third, the metadata contained subject ID, time point, age group and cohort information, which allowed the construction of a clean subject-by-feature-by-time tensor. In the final analysis, I kept subjects with complete transcriptomic measurements at days 0, 1, 3 and 7. This gave 72 complete subjects: 31 young and 41 elderly.

3 Methods

3.1 Tensor-based methods

The main idea of tensorOmics is to analyse longitudinal omics data without losing the time structure. Instead of converting all samples into one flat matrix, the data are organised as a tensor, where each subject is represented by a complete trajectory.

In this report, I used two methods from the tensorOmics framework. The first one was tPCA, an exploratory method used to summarize the main sources of variation in a longitudinal dataset. The second one was tPLS-DA, a supervised method used when individuals are divided into known groups, such as short-lived and long-lived fish. In both cases, the important point is that the subject, rather than the isolated sample, remains the unit of analysis.

The analyses were carried out with the tensorOmics R package, following the general structure of the examples provided by the authors [5]. For comparison, I also used classical matrix-based PLS-DA with the mixOmics framework [6].

3.2 Fish aging analysis

For the first analysis, I used the processed RNA-seq data from GSE150318. Lowly expressed genes were removed, and the 1000 most variable genes were selected to reduce the size of the dataset while keeping the most informative part of the transcriptome.

The data were reorganized into the following tensor:

$$113 \times 1000 \times 2$$

The three dimensions corresponded to fish, genes and time points. Since the response variable was categorical, short-lived or long-lived, I used tPLS-DA as the main tensor-based method. I also applied matrix-based PCA and PLS-DA as reference analyses. The objective was not to build a final lifespan prediction model, but to test whether the longitudinal representation could reveal a lifespan-related transcriptomic signal.

After the first score plots, I added several checks to make the result more interpretable. I tested the first two tPLS-DA components with MANOVA, used a label permutation test, and

tested whether the components were related to the real age at death. I also compared the top loading genes with genes showing the largest expression change between 10 and 20 weeks.

3.3 Vaccine response analysis

For the second analysis, I used the processed series matrix from GSE79396. The sample names and metadata were parsed to extract subject ID, age group, age, sex, cohort and day after vaccination. I kept the four early time points D0, D1, D3 and D7, because they were available for most subjects and correspond to the early molecular response after vaccination.

Only subjects with complete measurements at all four time points were retained. This resulted in a balanced longitudinal tensor:

$$72 \times 2000 \times 4$$

The first dimension corresponded to subjects, the second to the 2000 most variable probes, and the third to time. Expression values were centered and scaled before tensor construction.

I then performed two exploratory analyses. First, I used classical matrix PCA, where each sample-time observation was treated as one point. Second, I used tPCA, where each point represented one subject summarized by the full D0–D1–D3–D7 transcriptomic trajectory. To evaluate whether the first two tPCA components were associated with age group, I used MANOVA on the component scores and a permutation version of the same test. I also tested each component separately with a linear model.

To make the second analysis biologically interpretable, I annotated representative top loading probes from the age-associated tPCA component using the GEO platform annotation GPL13158. I then compared these genes with the biological interpretation proposed in the original Cell article, especially the age-related inflammatory and innate immune network.

4 Results

4.1 Fish aging transcriptomic trajectories

I started with PCA mainly to get a feeling for the data before looking for any discriminant signal. In this view, the two lifespan groups did not form two clear clouds. Short-lived and long-lived fish were mostly intermingled, which means that the largest sources of variation in the expression data were not simply the future lifespan classes. I do not see this as a surprising result. Lifespan is a complex outcome, and at 10 or 20 weeks of age it would be unrealistic to expect one dominant transcriptomic direction to separate the animals cleanly.

The matrix-based PLS-DA changed the picture, but only partly. Because PLS-DA uses the lifespan group during the analysis, a clearer separation was expected. The plot showed more structure than PCA, especially along the discriminant direction. However, this result also has to be interpreted carefully. In this matrix version, the 10-week and 20-week samples are handled as separate observations, even when they come from the same fish. In other words, the model can use the group labels, but it does not really keep the idea that each fish has its own short longitudinal history.

This is why the tensor-based tPLS-DA was the most meaningful analysis for this dataset. Here, each point corresponds to one fish, and the two measurements are interpreted together as a trajectory from 10 to 20 weeks. The separation between short-lived and long-lived fish was still not complete, but it became easier to read biologically: the first component seemed to

capture part of the difference between the two lifespan groups. I would therefore describe this result as a moderate signal rather than a strong classifier. It suggests that early transcriptomic trajectories contain some lifespan-related information, but not enough to make the phenotype look simple.

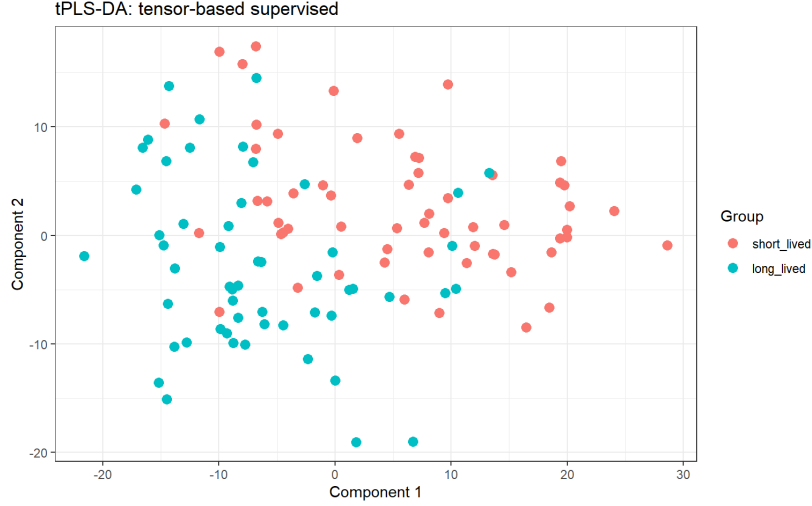


Figure 1: Tensor-based tPLS-DA of the fish aging RNA-seq dataset. Each point represents one fish trajectory.

The additional statistical checks supported this interpretation without changing the cautious conclusion (Table 1). MANOVA on the first two tPLS-DA components was significant for lifespan group, and the label permutation test was also significant. More importantly, the same components were associated with the real age at death. A linear model using Component 1 and Component 2 explained about 52% of the variation in lifespan. Component 1 was negatively correlated with lifespan ($r = -0.638$, $p = 2.90 \times 10^{-14}$), and Component 2 was also significant ($r = -0.335$, $p = 2.83 \times 10^{-4}$).

Table 1: Statistical follow-up of the fish aging tPLS-DA analysis.

Analysis	Statistic	p-value
tPLS-DA Comp1–Comp2 vs lifespan group	Pillai = 0.446	7.72×10^{-15}
tPLS-DA label permutation	Pillai = 0.446	0.0348
Age at death $\sim$ tPLS-DA Comp1 + Comp2	$R^2 = 0.520$	3.06×10^{-18}

The loading analysis helped me ask a more practical question: which variables were driving this partial separation? Some probes contributed more strongly to the discriminant components, including anonymous gene IDs such as `Nfu_g_1_022498`. I also compared the top tPLS-DA loading genes with genes ranked by the 20-week minus 10-week expression change. Among the top 50 genes in each list, only three genes overlapped, which was close to random expectation (expected overlap 2.5; hypergeometric $p = 0.463$). This suggests that the tPLS-DA result was not only a list of genes with the largest individual change.

The trajectories of representative top loading genes are shown in Figure 2. In several probes, the two lifespan groups followed visibly different 10-to-20-week profiles. Some genes remained higher in long-lived fish, while others showed a stronger decrease in short-lived fish.

This helped connect the score-level separation to actual longitudinal expression patterns. However, because many of these features are still anonymous killifish gene IDs, I treated them as candidate discriminant features rather than as final mechanistic markers.

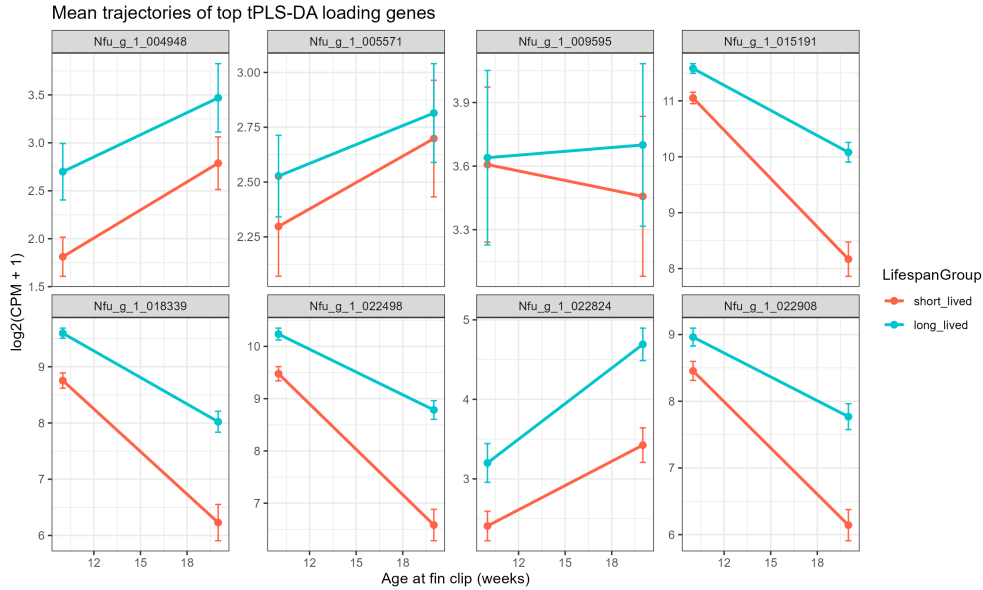


Figure 2: Mean trajectories of selected top tPLS-DA loading genes in the fish aging dataset. The plot shows how some discriminant features changed differently between short-lived and long-lived fish from 10 to 20 weeks of age.

4.2 Construction of the vaccine response tensor

For the second dataset, my aim was not only to repeat the same workflow on another example, but to check whether the tensor idea still made sense in a different biological context. From GSE79396, I kept the subjects who had complete PBMC transcriptomic measurements at D0, D1, D3 and D7. This produced a tensor with 72 subjects, 2000 probes and 4 time points. The two age groups were not perfectly balanced, but both were well represented, with 31 young and 41 elderly subjects.

The classical matrix PCA in Figure 3A was useful as a first diagnostic plot. It shows the global spread of all sample-time observations, but it is not very easy to read as a longitudinal result. The same subject can appear four times, once for each day, and the plot mixes several possible sources of variation: age group, sampling day, cohort and individual differences. This makes the figure informative, but also somewhat crowded in terms of interpretation.

The tPCA view in Figure 3B answers a slightly different question. Instead of asking where each sample-time point falls, it asks where each subject falls when the whole early response trajectory is considered together. This representation felt more natural for the biological question, because vaccination response is not just one isolated measurement. In this plot, the first component appeared to capture a strong cohort-related structure, while the second component showed a clearer age-associated pattern. That distinction was important: the strongest variation in the data was not necessarily the biological contrast I was most interested in, but the age signal was still visible once the components were examined more carefully.

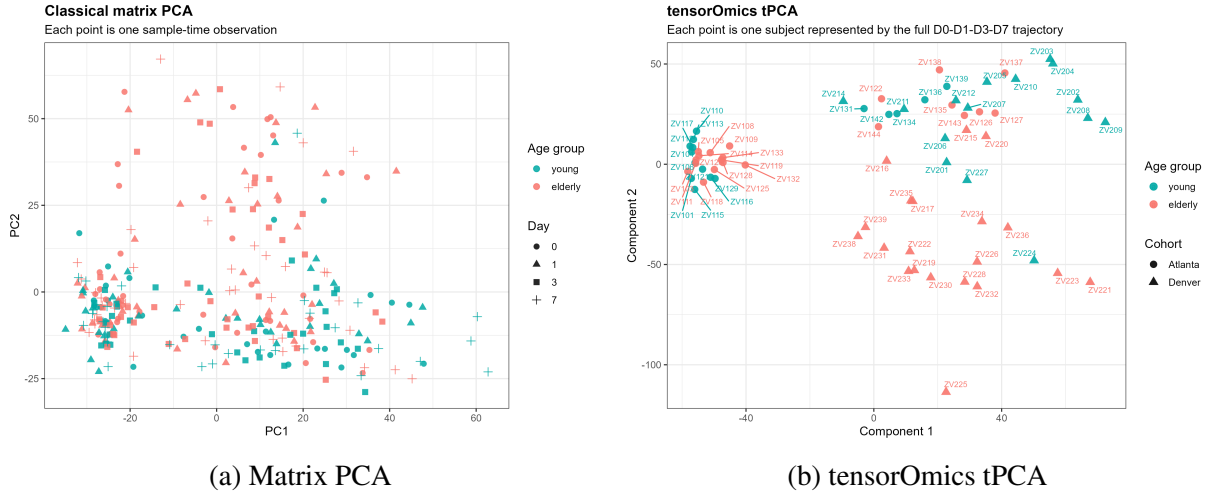


Figure 3: Comparison between matrix PCA and tPCA on the GSE79396 vaccine dataset. In matrix PCA, each point is a sample-time observation. In tPCA, each point is one subject represented by the full early post-vaccination trajectory.

4.3 Age-associated signal in tPCA

To check whether this visual impression was supported statistically, I tested the tPCA scores against age group. When the first two tPCA components were considered together, the association was significant: MANOVA gave a Pillai statistic of 0.186 with a p-value of 8.09×10^{-4} , and the permutation version also remained significant ($p = 0.002$). Looking at the two components separately made the result easier to interpret. Component 1 was not associated with age group ($p = 0.932$), whereas Component 2 was strongly associated with age group ($p = 1.53 \times 10^{-4}$).

Table 2: Statistical tests for the GSE79396 vaccine response analysis.

Analysis	Test	p-value
tPCA Comp1–Comp2 vs age group	MANOVA Pillai	8.09×10^{-4}
tPCA Comp1–Comp2 vs age group	Permutation MANOVA	0.002
Subject-averaged matrix PCA PC1–PC2 vs age group	MANOVA Pillai	9.72×10^{-4}
Subject-averaged matrix PCA PC1–PC2 vs age group	Permutation MANOVA	0.004
tPCA Component 1 vs age group	Linear model	0.932
tPCA Component 2 vs age group	Linear model	1.53×10^{-4}

These numbers should not be read as a simple competition where tPCA wins because it gives a smaller p-value. The subject-averaged matrix PCA was also significant. The more interesting point is what each representation considers to be the object of analysis. Matrix PCA either treats time-point samples as independent or requires averaging afterward. tPCA keeps the subject-level trajectory intact from the beginning. For this dataset, that made the age-related pattern easier to connect with the design of the experiment.

The probes with the strongest negative loadings on Component 2 gave a more concrete view of this pattern. Across the four time points, elderly subjects tended to have higher scaled

expression than young subjects for these top probes. Several trajectories also showed an early change around day 1, followed by lower values at days 3 and 7. This suggests that Component 2 is not just a statistical axis: it reflects a time-dependent transcriptional pattern that differs between young and elderly subjects during the early vaccine response.

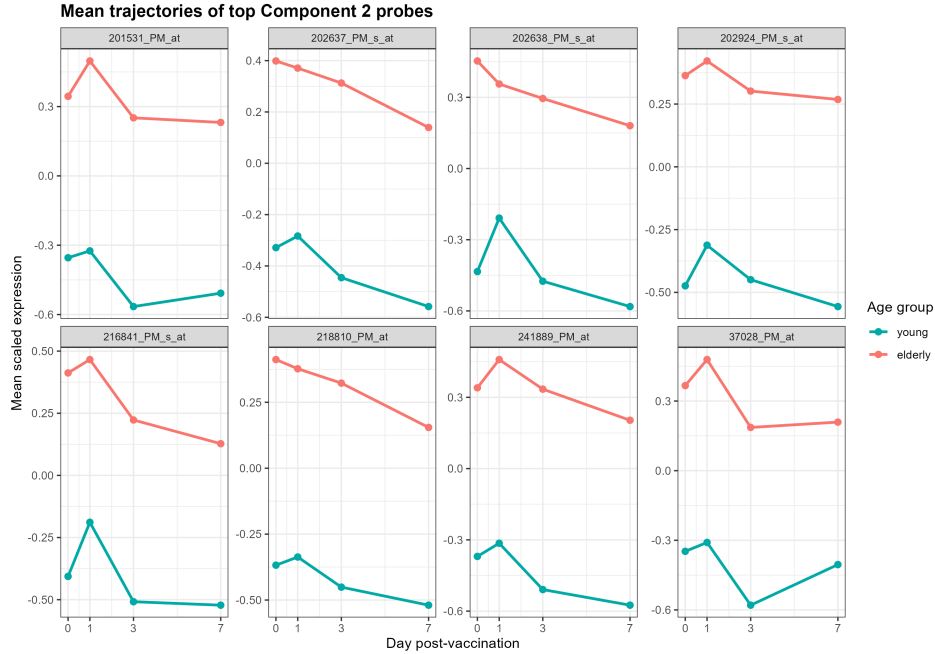


Figure 4: Mean trajectories of the top probes contributing to tPCA Component 2 in GSE79396. Component 2 was significantly associated with age group.

This result also had a biological interpretation. After annotation, several top Component 2 genes were consistent with the age-related inflammatory network described in the original Cell article. Li et al. reported that elderly subjects showed stronger baseline inflammatory and innate immune signatures. Some examples among the annotated tPCA genes are shown in Table 3. Therefore, the tensor result was not only a statistical separation between young and elderly subjects. It also recovered a trajectory-level view of an inflammatory and stress-response state that had already been described by the original network analysis.

Table 3: Examples of annotated genes among the top tPCA Component 2 loadings.

Gene	Biological interpretation
<i>IL1B</i>	Pro-inflammatory cytokine involved in innate immune activation.
<i>NLRP3</i>	Inflammasome-related gene linked to inflammatory response.
<i>ICAM1</i>	Cell adhesion molecule involved in leukocyte interaction and inflammation.
<i>SOD2</i>	Mitochondrial oxidative stress response gene.
<i>NFKBID</i>	Regulator of the NF- κ B pathway and inflammatory signaling.

5 Discussion

5.1 Comparison between the two applications

The two datasets did not behave in the same way, and this difference was useful for understanding what the methods were doing. In the fish dataset, the target phenotype was future lifespan, inferred from early transcriptomic profiles. This is a difficult and indirect question. Lifespan depends on many processes, and many of them may not be visible as a single strong transcriptomic direction at the time points measured here. Therefore, the partial separation obtained with tPLS-DA is best interpreted as a limited but interesting signal. The statistical follow-up strengthened this interpretation because the components were also associated with real age at death, not only with the binary lifespan group.

The vaccine dataset was different. The contrast between young and elderly adults is more directly connected to immune function, and early transcriptional changes after vaccination are biologically expected. In that setting, tPCA found an age-associated component, mainly Component 2. What I found convincing was not only that the statistical tests were significant, but that the component could be interpreted as a subject-level trajectory and was linked to inflammatory and innate immune genes. This is where the tensor representation felt useful: it allowed the analysis to keep the temporal structure instead of reconstructing it afterward.

5.2 Why GSE79396 was a useful second dataset

In the tensorOmics manuscript, the FMT:HIV case study was used to demonstrate block.tPLS-DA in a longitudinal multi-omic setting. In that example, the method integrated microbial taxonomy, microbial functions and host proteins, and it was able to separate FMT recipients from placebo controls while highlighting treatment-related multi-omic signatures. This showed that tensorOmics could be useful when several biological layers are measured over time in the same subjects.

For my second analysis, I wanted to test a related idea in a different context. If the biological system, the type of measurements and the metadata are different, can a tensor-based method still give an interpretable result? I therefore selected GSE79396. This dataset is not identical to the FMT case study, because I used a single-omic tPCA analysis rather than multi-block integration. However, it shares an important feature with the FMT example: it is a human longitudinal intervention dataset, with repeated measurements collected from the same subjects after vaccination.

There was also a practical reason for this choice. The sample names and metadata were clear enough to rebuild the longitudinal design, most subjects had complete early time points, and the young-versus-elderly comparison was easy to formulate biologically. The final tensor was large enough to be meaningful, but still small enough to analyze on a standard computer. This made GSE79396 a good compromise between biological interest, methodological relevance and technical feasibility.

5.3 Methodological limitations

Several points limit how far the results can be interpreted. The first one is cohort structure. The vaccine dataset includes samples from Atlanta and Denver, and Component 1 in the tPCA plot appeared strongly related to cohort. I do not see this as a failure of tPCA. On the contrary, it shows that the method captures strong structured variation in the tensor, whether that variation

is biological, technical or both. But it also means that the age-related interpretation should focus mainly on Component 2, and cohort should be kept in mind as a possible confounding factor.

A second limitation comes from feature selection. I used the 2000 most variable probes to make the analysis more stable and easier to visualize. This is reasonable for exploration, but it also means that some lower-variance genes with biological relevance may have been excluded. In addition, I only annotated a small set of representative top genes. Since I did not perform a complete pathway enrichment analysis, the biological interpretation remains a focused comparison with the original article rather than a full mechanistic study.

Finally, tPCA itself is unsupervised. It can reveal axes that are associated with age group, but it is not designed to classify subjects or optimize prediction of vaccine response. I also explored a response-level tPLS analysis using D30/D0 IgG and D7/D0 CXCR3-positive Tfh response as outcomes. The fitted components were correlated with both responses, but the permutation tests and cross-validation did not support a robust predictive model. For this reason, I did not keep this analysis as a main result. It is more appropriate to mention it as a possible extension.

5.4 Perspectives

The most useful next step would be to perform a formal pathway enrichment analysis on the probes contributing most strongly to Component 2. This could test more systematically whether the age-associated trajectory is connected to known vaccine response processes, such as interferon signaling, innate immune activation, inflammation, antigen presentation or B cell response. Another extension would be to include antibody titers or immune cell responses as response variables. This would move the analysis from describing an age-associated transcriptional trajectory toward asking whether early molecular changes are related to vaccine immunity.

6 Conclusion

Overall, this project helped me understand tensorOmics less as just another dimensionality reduction tool and more as a way to preserve the structure of longitudinal omics data. In the fish aging dataset, tPLS-DA revealed a partial separation between short-lived and long-lived individuals, and the same components were associated with real age at death. In the independent vaccine dataset GSE79396, tPCA identified an age-associated transcriptional trajectory after Herpes zoster vaccination. Annotation of representative top genes suggested that this trajectory was related to inflammatory and innate immune processes described in the original Cell article. The main value of the tensor approach was therefore not only statistical significance, but the way it kept each subject as a complete temporal trajectory. For exploratory longitudinal omics analysis, this makes the results easier to connect back to the biological design.

Use of artificial intelligence tools

Artificial intelligence tools were used to help with language correction and report organization. The data analysis, results, figures and scientific interpretations were checked and validated by myself.

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